

MUSTANG BEACH, TX, USA

WMO: 722008

Lat: 27.812N

Lon: 97.089W

Elev: 2

StdP: 101.31

Time Zone: -6.00 (NAC)

Period: 05-19

WBAN: 12995

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a) 1	(b) 2.6	(c) 4.8	(d) -4.4	(e) 2.6	(f) 7.9	(g) -2.0	(h) 3.2	(i) 8.9	(j) 12.7	(k) 13.5	(l) 11.7	(m) 11.4	(n) 7.1	(o) 0	(p) 0.478

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a) 8	(b) 4.5	(c) 32.6	(d) 27.0	(e) 32.3	(f) 27.0	(g) 32.0	(h) 27.0	(i) 28.9	(j) 30.4	(k) 28.3	(l) 30.2	(m) 27.9	(n) 30.0	(o) 5.0	(p) 150

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
28.7	25.2	29.5	27.8	23.8	29.0	27.4	23.2	28.9	94.7	29.2	90.6	29.8	89.3	30.6	31.2

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a) 11.0	(b) 9.5	(c) 8.5	(e) 0.9	(f) 34.2	(g) 1.7	(h) 1.8	(i) -0.3	(j) 35.5	(k) -1.3	(l) 36.6	(m) -2.3	(n) 37.6	(o) -3.5	(p) 38.9		
(a) 11.0	(b) 9.5	(c) 8.5	(e) -1.3	(f) 28.9	(g) 1.9	(h) 1.0	(i) -2.7	(j) 29.6	(k) -3.8	(l) 30.2	(m) -4.9	(n) 30.8	(o) -6.3	(p) 31.5		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	22.7	13.8	15.6	18.8	22.7	25.7	28.7	29.2	29.5	28.4	25.1	19.3	15.6
	DBStd	6.48	4.26	4.60	3.75	2.75	2.14	1.20	0.97	1.03	1.92	3.22	4.84	4.41
	HDD10.0	44	17	12	2	0	0	0	0	0	0	0	4	8
	HDD18.3	439	147	93	39	5	0	0	0	0	0	3	46	107
	CDD10.0	4688	135	168	276	380	486	560	594	606	551	468	283	181
	CDD18.3	2042	6	15	54	135	228	310	336	347	301	212	76	21
	CDH23.3	20734	3	14	57	532	1910	3725	4212	4475	3569	1928	297	12
	CDH26.7	7499	0	2	4	56	342	1419	1767	2023	1429	439	18	0
Wind	WSAvg	4.7	4.5	4.5	4.7	5.3	5.4	5.0	4.7	4.2	4.4	4.5	4.5	4.3
Precipitation	PrecAvg	870	53	53	66	53	83	67	80	50	150	106	71	42
	PrecMax	1269	159	135	137	219	315	225	351	183	421	357	253	153
	PrecMin	462	0	2	2	5	0	0	0	1	33	0	0	8
	PrecStd	241	45	44	42	52	85	56	100	46	94	99	68	39
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	23.0	25.5	27.0	31.1	31.0	32.5	32.7	33.0	34.0	31.5	28.1	24.8
		MCWB	17.9	18.3	21.4	18.1	25.5	26.2	27.5	27.5	25.7	25.8	24.2	20.1
	2%	DB	22.0	22.8	24.6	27.8	29.8	32.1	32.3	32.7	32.6	30.7	27.2	23.0
		MCWB	18.7	19.9	21.9	23.6	26.0	26.6	27.4	27.4	26.6	25.8	23.8	21.0
	5%	DB	20.9	22.0	23.7	27.1	29.0	31.3	32.0	32.3	32.1	29.7	26.2	22.4
		MCWB	17.9	19.5	21.6	23.9	25.7	26.6	27.3	27.3	26.7	25.1	23.1	20.5
	10%	DB	19.4	21.0	22.8	26.1	28.5	31.0	31.3	32.0	31.3	28.8	25.1	21.6
		MCWB	17.6	18.9	21.0	23.3	25.5	26.5	26.9	27.3	26.3	24.6	22.6	19.9
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	21.7	21.9	24.2	27.3	28.9	29.4	29.6	28.8	29.8	28.5	26.0	22.9
		MCDB	22.1	23.1	24.5	27.4	29.0	30.3	30.6	31.5	30.8	29.2	27.1	23.5
	2%	WB	20.1	21.1	23.0	26.1	27.5	28.0	28.9	28.4	28.5	27.4	25.0	22.2
		MCDB	20.7	22.1	23.7	26.6	27.9	29.7	30.3	31.1	30.1	28.4	26.3	22.8
	5%	WB	19.0	20.3	22.5	25.1	27.0	27.7	28.1	28.0	27.7	26.5	24.0	21.2
		MCDB	19.8	21.2	23.2	26.2	27.8	29.5	30.0	30.8	30.0	28.3	25.3	21.9
	10%	WB	18.0	19.4	21.9	24.1	26.3	27.3	27.7	27.7	27.2	25.7	23.1	20.5
		MCDB	18.7	20.2	22.6	25.2	27.6	29.3	29.9	30.5	29.8	28.2	24.7	21.2
Mean Daily Temperature Range	5% DB	MDBR	6.5	6.4	5.4	5.1	4.5	4.2	4.3	4.5	5.0	5.6	6.2	6.6
		MCDBR	7.4	6.6	5.1	5.6	4.6	4.5	4.5	4.8	5.3	5.0	5.3	6.4
		MCWBR	6.1	4.9	3.7	3.9	2.4	2.0	1.9	1.9	2.5	3.0	3.8	5.3
	5% WB	MCDBR	6.4	5.8	4.5	4.5	4.1	4.1	4.4	4.5	5.0	4.9	5.7	6.2
		MCWBR	6.1	5.0	3.9	3.4	2.8	2.2	2.2	1.9	2.8	4.2	5.0	5.8
Clear-Sky Solar Irradiance	taub	0.363	0.370	0.388	0.433	0.474	0.450	0.468	0.452	0.440	0.394	0.379	0.353	
	taud	2.482	2.465	2.412	2.290	2.208	2.341	2.307	2.350	2.343	2.471	2.463	2.528	
	Ebn at Noon	879	902	904	868	828	843	827	840	841	866	853	874	
	Edh at Noon	95	104	116	134	146	127	131	125	122	102	96	87	
All-Sky Solar Radiation	RadAvg	2.99	3.55	4.56	5.44	6.13	6.72	6.61	6.32	5.22	4.56	3.41	2.69	
	RadStd	0.33	0.58	0.48	0.52	0.42	0.35	0.55	0.34	0.44	0.46	0.36	0.36	

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)
Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
Regional (26 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	+81

Nomenclature: See separate page